

Final project

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2026-02-28

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Introduction

This report follows the required steps for a longitudinal social network analysis using stochastic actor-oriented models (SAOMs) in **RSiena**. We analyze directed “*best friends*” nominations collected in Dutch secondary school classrooms across multiple waves (waves **V**, **X**, **Y**).

Data source. The classroom network and pupil attributes come from the dataset collected by Andrea Knecht as part of her dissertation research (early adolescence, 2003/04) and distributed via the course materials/DANS archive:

- Knecht, A. (2006), *Networks and actor attributes in early adolescence* [2003/04], Utrecht University / ICS (Codebook no. 61).

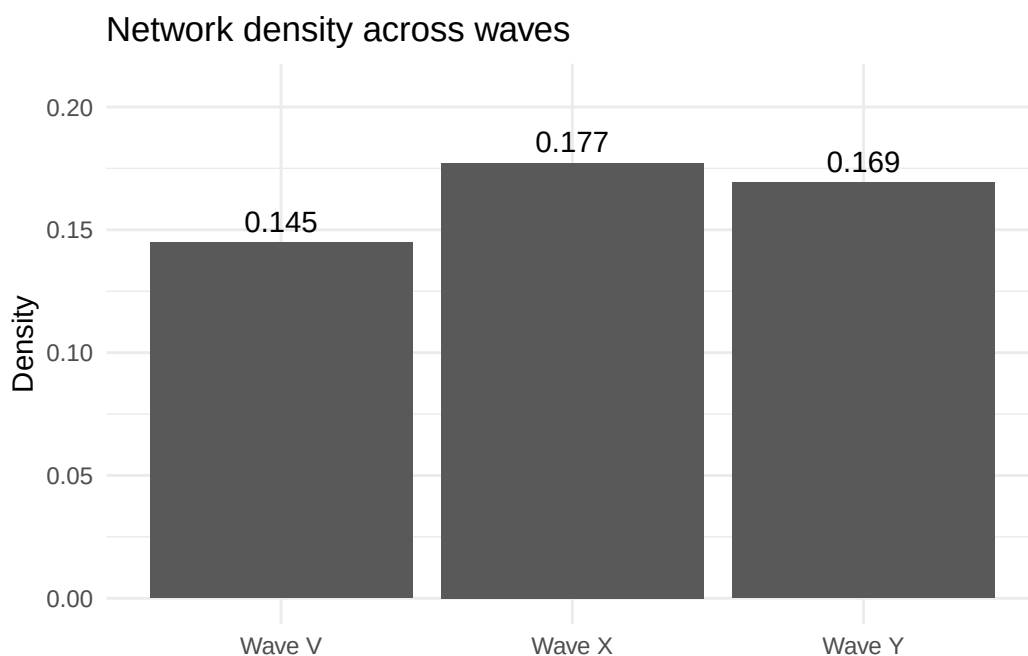
Research focus. We examine whether (i) **sex homophily** shapes best-friend nominations (dyadic hypothesis), (ii) **triadic closure/transitivity** drives clustering (triadic hypothesis), and (iii) a **node-level attribute** (sex-based *activity*) predicts how many best-friend nominations pupils send (ego/actor hypothesis). We also include standard endogenous network controls (e.g., reciprocity and degree-related effects) and evaluate model fit using goodness-of-fit (GoF) simulations.

1. Network Descriptives

We focus on classroom **09f** (variable `schoolnr`). This classroom was selected because it provides at least three observed waves and shows sufficient stability (composition and tie Jaccard) to support SAOM estimation while still exhibiting meaningful change over time.

We generate a figure showing network density across waves.

Figure 1: Network density across waves



These values are proportions of possible friendships that exist. So for Wave V, only about 14.5% of all possible ties were actually reported. Wave X is slightly denser (about 17.7%), and Wave Y is very similar (about 16.9%). Networks with density < 0.2 are considered sparse, which is very common in social networks because

not everyone is friends with everyone else. We generate a table showing basic descriptives of the best-friends network across waves.

Table 1: Basic descriptives of the best-friends network across waves

Wave	Nodes	Density	DensityChangePct
Wave V	30	0.145	NA
Wave X	28	0.177	22.386
Wave Y	26	0.169	-4.524

For classroom **09f** (variable `schoolnr`), over the three waves of data collection the network shows a moderate change in its social structure. The number of students decreased slightly from 30 in Wave V to 28 in Wave X, and then to 26 in Wave Y, indicating some turnover in classroom composition. The density of the “best friends” network increased from 0.145 in Wave V to 0.177 in Wave X (a 22.4% increase), before changing to 0.169 in Wave Y (a -4.5% change relative to Wave X). Overall, the classroom exhibits a relatively stable friendship structure, which is suitable for longitudinal network modelling.

For the selected classroom, the “Best friends” network was constructed from the survey responses, transforming each student’s nominations into an adjacency matrix. Missing codes 98/99 (as documented in the codebook) were treated as missing nominations (i.e., no tie created).

We generate tables showing (i) the Jaccard stability of the node set across waves and (ii) the number of students present in all three waves.

Table 2: Jaccard stability of the node set (classroom composition)

Wave_Comparison	Jaccard_NodeSet
Wave V $\rightarrow$ Wave X	0.933
Wave X $\rightarrow$ Wave Y	0.862

Table 3: Number of students present in all three waves (intersection)

Students_in_all_three_waves
25

Table 4: Jaccard stability of ties on the constant node set (intersection sample)

Wave_Comparison	Jaccard_Ties
Wave V $\rightarrow$ Wave X	0.242
Wave X $\rightarrow$ Wave Y	0.470

The table above reports **tie Jaccard** stability (overlapping friendships) between consecutive waves on the constant node set used for SAOM. Higher values indicate more stable friendship relations across time, supporting the use of longitudinal modelling.

The Jaccard indices above quantify how stable the classroom composition is across waves. Because SAOMs require a constant node set, the subsequent SAOM analysis will be run on the students who are present in all three waves (the intersection).

2. Dyadic Hypothesis

For the analysis of dyadic characteristics in the “Best friends” network, I formulated the following hypothesis: students are more likely to nominate as best friends those peers who are of the same sex. This hypothesis is based on the well-established principle of sex homophily in adolescent social networks, where boys tend to form friendships with other boys, and girls with other girls. Theoretical considerations, such as social identity theory, suggest that individuals prefer closer ties with peers who share salient characteristics like sex, which can also facilitate participation in similar classroom activities or interests. Empirical studies in classroom social networks consistently report high levels of sex-based homophily (e.g., Snijders et al., 2010). To operationalize this dyadic hypothesis, a sex similarity matrix was constructed for each wave, where each cell represents whether a pair of students share the same sex (coded as 1) or not (coded as 0). This matrix can later be used in stochastic actor-oriented models (SAOMs) to formally test the influence of sex similarity on the formation and maintenance of friendship ties over time. We generate tables showing 5×5 subsets of the sex similarity matrices for each wave.

Table 5: Sex similarity matrix (Wave V; 5×5 subset)

1	2	3	4	5
1	1	0	1	0
1	1	0	1	0
0	0	1	0	1
1	1	0	1	0
0	0	1	0	1

Table 6: Sex similarity matrix (Wave X; 5×5 subset)

1	2	3	4	5
1	0	1	0	1
0	1	0	1	0
1	0	1	0	1
0	1	0	1	0
1	0	1	0	1

Table 7: Sex similarity matrix (Wave Y; 5×5 subset)

	1	2	3	4	6
1	1	1	0	1	1
2	1	1	0	1	1
3	0	0	1	0	0
4	1	1	0	1	1
6	1	1	0	1	1

To operationalize this dyadic hypothesis, sex similarity matrices were constructed for each wave, where each cell represents whether a pair of students share the same sex (coded as 1) or not (coded as 0). Examples of these matrices for a subset of students are:

Wave V (subset 5×5) 1 2 3 4 5 1 1 1 0 1 0 2 1 1 0 1 0 3 0 0 1 0 1 4 1 1 0 1 0 5 0 0 1 0 1

Wave X (subset 5×5): 1 2 3 4 5 1 1 0 1 0 1 2 0 1 0 1 0 3 1 0 1 0 1 4 0 1 0 1 0 5 1 0 1 0 1

Wave Y (subset 5×5, new student 6 included): 1 2 3 4 6 1 1 1 0 1 1 2 1 1 0 1 1 3 0 0 1 0 0 4 1 1 0 1 1 6 1 1 0 1 1

These matrices can be used in stochastic actor-oriented models (SAOMs) to formally test the influence of sex similarity on the formation and maintenance of friendship ties across waves. The consistent pattern of higher values along clusters of same-sex students supports the theoretical expectation that sex homophily plays a role in the friendship formation in this classroom

3. Triadic Hypothesis

For the analysis of triadic characteristics in the “Best friends” network, I formulated the following hypothesis: friendship ties are likely to exhibit transitivity, meaning

that if student A nominates student B as a best friend, and student B nominates student C, then student A is also likely to nominate student C. This hypothesis is grounded in the principle of triadic closure, a common phenomenon in social networks, where “friends of friends tend to become friends” (Granovetter, 1973; Snijders et al., 2010). Theoretical explanations suggest that transitive ties increase trust, reduce social tension, and facilitate cooperation within a group. In classroom networks, transitivity is often observed because students share activities, classes, and social environments, making it more likely that friendship clusters form.

To operationalize this triadic hypothesis, a transitivity measure or a triadic closure matrix can be calculated for each wave. Each cell in such a matrix represents whether a transitive relationship exists between a set of three students (coded as 1 if present, 0 otherwise). These matrices or transitivity statistics can later be included in stochastic actor-oriented models (SAOMs) to formally test whether the presence of a friendship between two students increases the likelihood of forming a friendship with a third connected student, capturing the clustering tendency in the classroom over time

We generate a table showing global transitivity (triadic closure) for each wave.

Table 8: Global transitivity by wave

Wave	Transitivity
Wave V	0.479
Wave X	0.610
Wave Y	0.500

The transitivity of the “Best friends” network, which measures the tendency of friends-of-friends to also be friends (triadic closure), shows a moderate to high level of clustering across the three waves. In Wave V, the transitivity is 0.479, indicating that a substantial share of possible triads are closed. Wave X exhibits a transitivity of 0.61, reflecting the tendency for friendship triads to form. In Wave Y, the transitivity is 0.5, suggesting that the level of triadic closure remains in a similar range over time. These results support the hypothesis that friendship formation in the classroom is characterized by transitive relationships, consistent with established findings in adolescent social networks where social groups tend to form tightly connected clusters

4. Node Attribute Hypothesis

Node-level attribute hypothesis (sex-based activity). Beyond *with whom* pupils befriend (homophily), pupils may differ in *how actively* they nominate best

friends. We therefore test the hypothesis that **sex predicts outgoing friendship nominations**: one sex is, on average, more likely to send best-friend nominations over time (higher outdegree), net of other structural tendencies. In SAOM terms, this is captured by an **ego effect** of sex (`egoX` with `sex_cov`).

Below, we provide a brief descriptive check of outdegree by sex (mean nominations sent) across waves.

Table 9: Mean outdegree (nominations sent) by sex and wave

Wave	sex	N_students	Mean_outdegree	SD_outdegree
Wave V	1	12	4.250	2.701
Wave V	2	18	4.167	3.167
Wave X	1	11	5.182	2.926
Wave X	2	17	4.529	2.625
Wave Y	1	10	3.800	0.632
Wave Y	2	15	4.533	2.200
Wave Y	3	1	4.000	NA

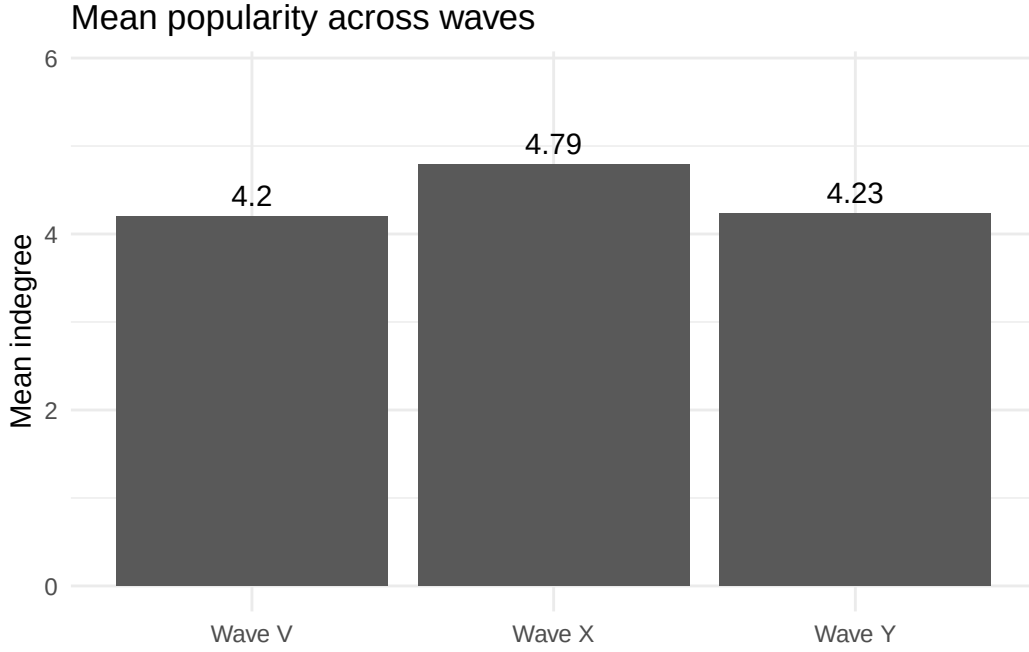
As an additional descriptive check, we also summarise **popularity (indegree = nominations received)** across waves. In the SAOM, the tendency for already-popular pupils to attract (or repel) additional nominations is represented by the endogenous **indegree-related popularity effect (inPop)**, which we treat as a *structural control* rather than the focal node-attribute hypothesis.

Table 10: Popularity (indegree) summary by wave

Wave	MeanPopularity	MinPopularity	MaxPopularity
Wave V	4.20	0	8
Wave X	4.79	1	8
Wave Y	4.23	1	7

For the node-level analysis, the popularity of each student was calculated as the total number of best-friend nominations they received in each wave. The data were initially transformed into a long format, where each student in each wave was represented as a separate row, resulting in 84 rows in total (30 students in Wave V, 28 in Wave X, and 26 in Wave Y). To summarize the key characteristics of this node-level variable, we computed the mean, minimum, and maximum popularity per wave. In Wave V, students received on average 4.2 nominations, with a range from 0 to 8. In Wave X, the mean popularity was 4.79, with a minimum of 1 and a maximum of 8. In Wave Y, the mean popularity was 4.23, with a minimum of 1

Figure 2: Mean popularity (indegree) across waves



and a maximum of 7. These summaries indicate that while most students receive several friendship nominations, some students are consistently more popular than others, which may influence the formation and maintenance of best-friend ties over time.

5. Control Variables

In addition to the three focal hypotheses (sex homophily, transitivity, and the sex-based ego effect), we include standard **endogenous network controls** to avoid attributing basic structural tendencies to covariates. Specifically, we control for (i) **baseline sparsity** (outdegree/density), (ii) **reciprocity**, (iii) **degree-related heterogeneity in receiving nominations** (indegree-related popularity, `inPop`), and (iv) **degree-related heterogeneity in sending nominations** (outdegree activity, `outAct`). Together with the covariate-based effects (`sameX` and `egoX`) and the triadic closure term (`transTrip`), these controls help isolate the mechanisms of interest in the evolution of the best-friends network.

We generate a table listing the control variables used in the analysis.

Table 11: Control variables to be included in the analysis

Variable	Type	Description
Outdegree (density)	Structural	Baseline tendency toward a sparse nomination network.
Reciprocity	Dyadic/structural	Tendency to reciprocate a best-friend nomination.
Sex homophily (sameX)	Dyadic (covariate-based)	Preference for same-sex best-friend nominations (dyadic covariate effect).
Sex ego effect (egoX)	Ego-level (actor covariate)	Difference in sending nominations by sex (node/actor attribute effect on outdegree).
Transitivity (transTrip)	Triadic/structural	Friends-of-friends closure: $A \rightarrow B$ and $B \rightarrow C$ increases $A \rightarrow C$.
Indegree-related popularity (inPop)	Alter-level / degree-related (endogenous)	Whether pupils with higher indegree (already nominated often) become more/less likely to receive additional nominations.
Outdegree activity (outAct)	Ego-level / degree-related (endogenous)	Heterogeneity in pupils' tendency to send nominations beyond what is explained by covariates.

6. Longitudinal Plots

We generate figures showing the evolution of the best-friends network across waves (V, X, Y), using **fixed node positions** and **node colors by sex**. Fixed positions make it possible to compare structural change over time without the layout itself moving the nodes around. We focus on three waves (V, X, Y) to follow the recommendation of using at least three waves while keeping a sufficiently stable node set for SAOM estimation. Including Wave W would further reduce the overlap of pupils observed across all waves (intersection), which would decrease sample size and potentially reduce estimation stability.

7. Model Specification

In this project, we test three hypotheses using a stochastic actor-oriented model (SAOM) in RSiena. We operationalize each hypothesis by selecting the corresponding RSiena effect(s). We generate a table showing how each hypothesis is translated into a SAOM effect. For each hypothesis, we select the corresponding RSiena effect that captures the proposed mechanism of tie formation and maintenance in the best-friends network.

Figure 3: Best-friends network in Wave V (fixed node positions; node color indicates sex).

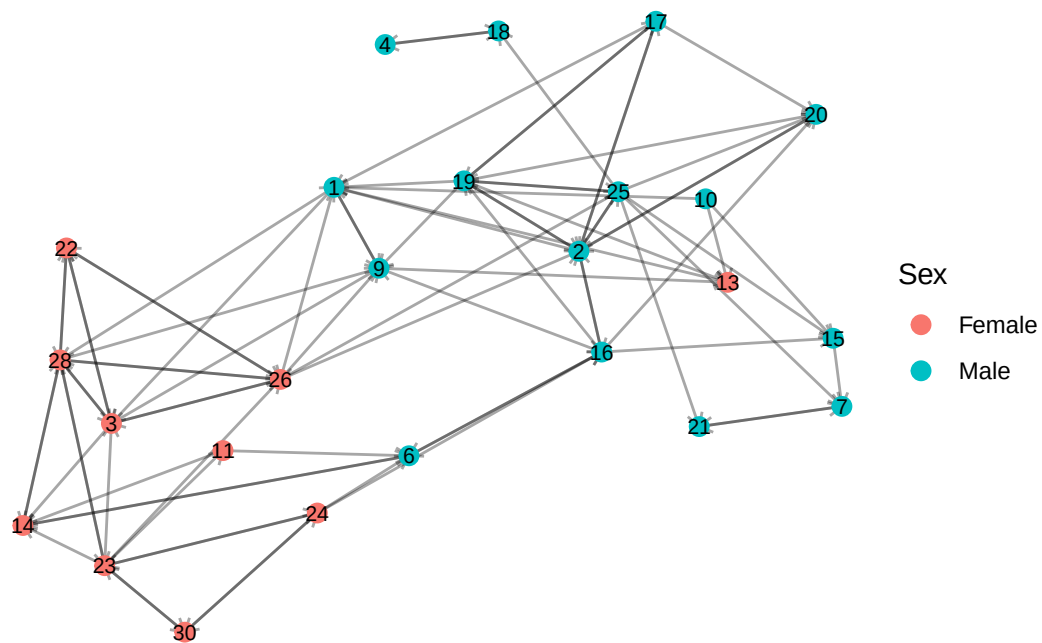


Figure 4: Best-friends network in Wave X (fixed node positions; node color indicates sex).

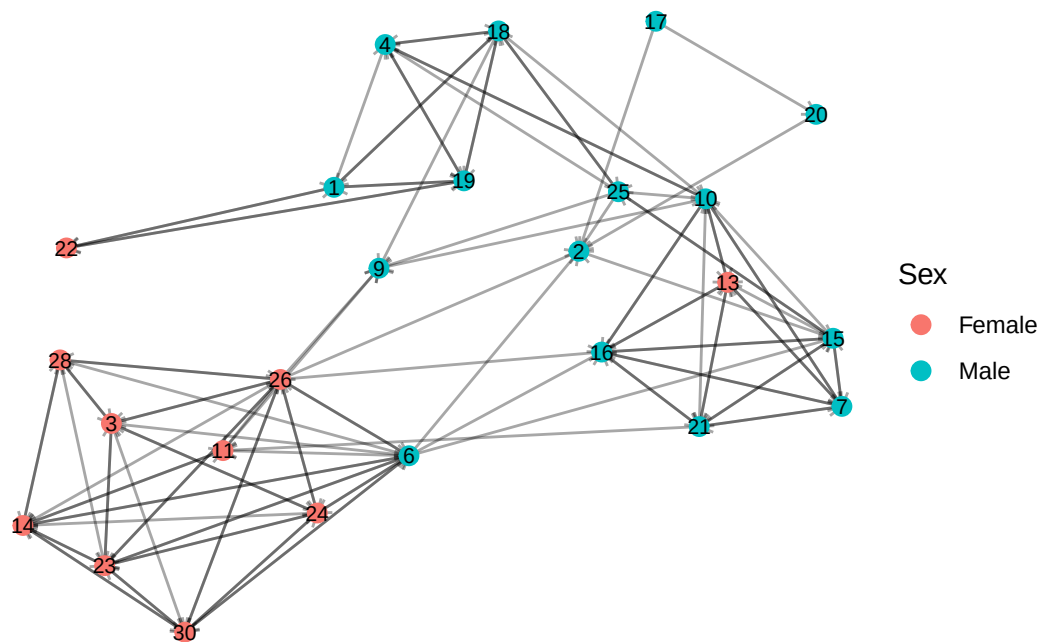


Figure 5: Best-friends network in Wave Y (fixed node positions; node color indicates sex).

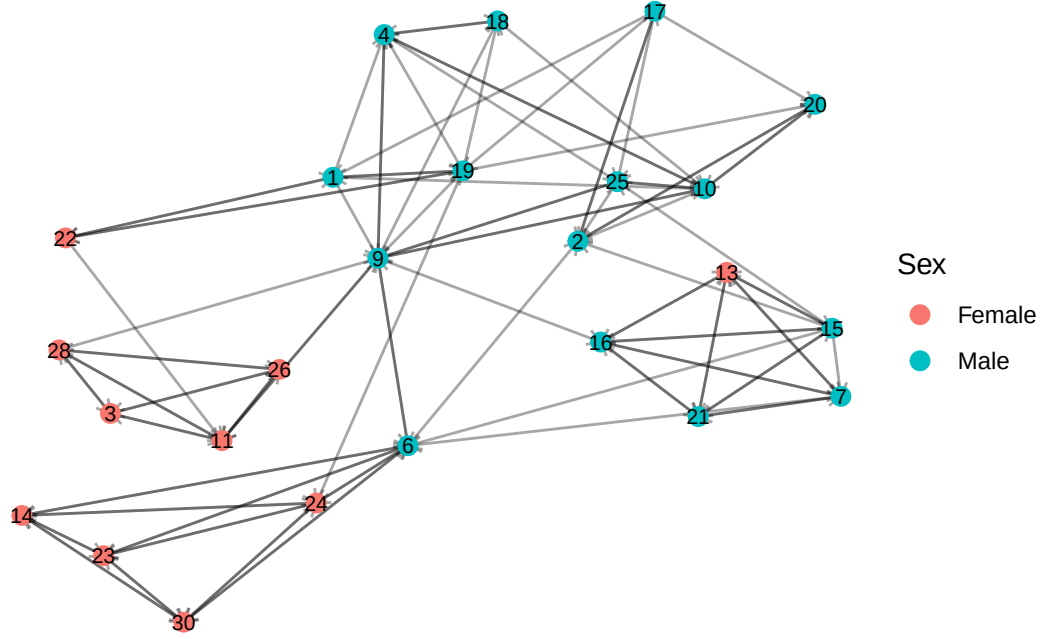


Table 12: Operationalization of hypotheses as RSiena effects.

Hypothesis	RSiena_effect	Expected_direction	Interpretation
H1 (Dyadic): sex homophily	sameX (interaction1 = sex_cov)	Positive	Ties are more likely between pupils of the same sex.
H2 (Triadic): transitivity	transTrip	Positive	Friend-of-a-friend closure: $A \rightarrow B$ and $B \rightarrow C$ increases $A \rightarrow C$.
H3 (Node-level): sex-based activity (outgoing nominations)	egoX (interaction1 = sex_cov)	Directional (to be tested)	One sex is more likely to send best-friend nominations over time (higher outdegree).

First, sex homophily is modeled with the sameX effect using the sex covariate (interaction1 = “sex_cov”). A positive parameter would indicate that pupils are more likely to nominate same-sex peers as best friends. Second, transitivity is captured by transTrip, where a positive parameter indicates triadic closure (friends-of-friends become friends). Third, popularity reinforcement is modeled with inPop (indegree popularity); a positive parameter indicates a Matthew effect in which pupils who already receive many nominations tend to attract additional nominations over time

8. Model Estimation

We conduct RSiena analyses to test the hypotheses above. Because SAOMs require the same node set across waves, we use the SAOM-aligned matrices (`netV_saom`, `netX_saom`, `netY_saom`) constructed from pupils present in all three waves. We estimate an initial model (Model 1) including our hypothesis-driven effects (`sameX`, `transTrip`, `inPop`) and a standard control for sender heterogeneity (`outAct`). If convergence is not reached, we increase estimation effort (`n3`) and continue from the previous estimates using `prevAns`.

We generate a table showing the estimated parameters of the final converged model.

Table 13: Convergence diagnostics for the final estimation run.

Metric	Value
tconv.max	0.091
max(tconv)	0.037

The Table 14 shows the parameter estimates for the final model (Estimate, standard error, and t-ratio).

Table 14: RSiena parameter estimates for the final model.

[illegible]

[illegible]

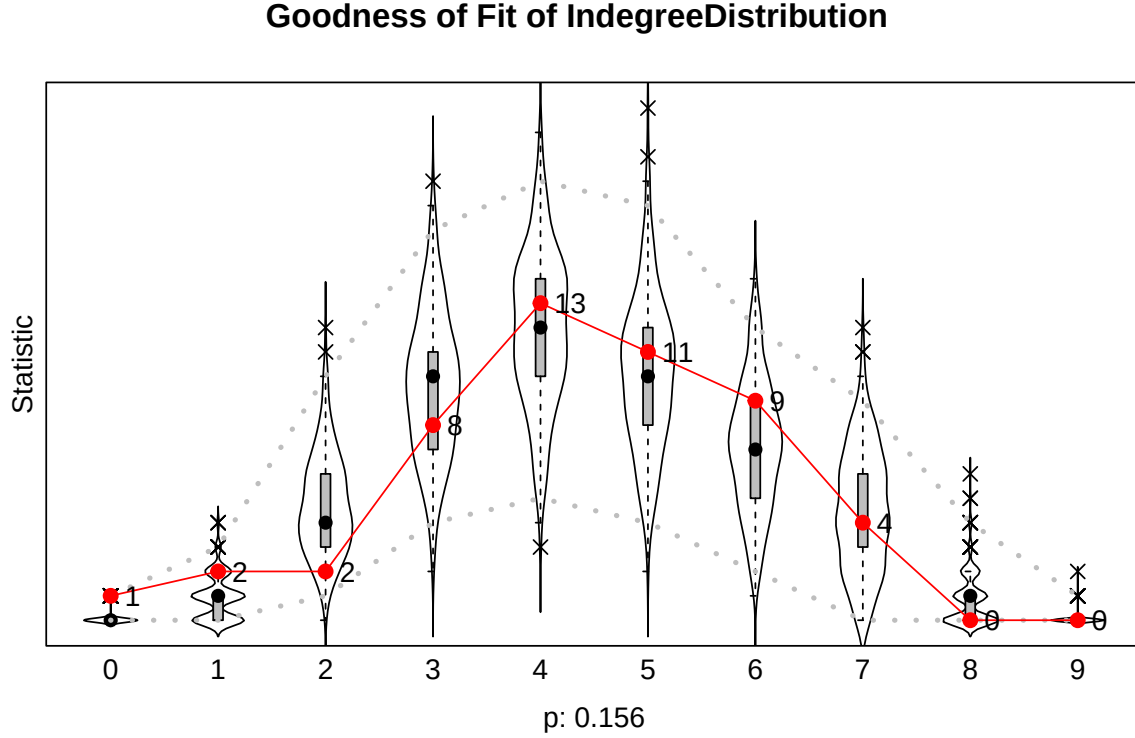
Convergence was achieved in the final run ($\text{tconv.max} = 0.091$ and $\max(|\text{tconv}|) = 0.037$), so we proceed with interpretation of the estimated parameters

9. Model Fit

Goodness-of-fit (GoF) assesses whether the fitted SAOM can reproduce important structural features of the observed network. We simulate multiple networks from the fitted model and compare observed network statistics to the distribution of the same statistics in the simulated networks. If the observed statistics are not extreme relative to the simulated distribution (and the Monte Carlo p-values are not close to 0 or 1), the model is considered to fit those features reasonably well. We evaluate GoF for the indegree distribution, outdegree distribution, and the triad census for our final model from Question 8 (Model 1). We then update the model once (Model 2) by adding an additional endogenous effect (**reciAct**) and re-run GoF to assess whether fit improves.

Note: some statistics are not plotted because their variance is 0.
This holds for the statistics: 10 11 12 13 14 15.

Figure 6: Goodness-of-fit for the indegree distribution (Model 1).



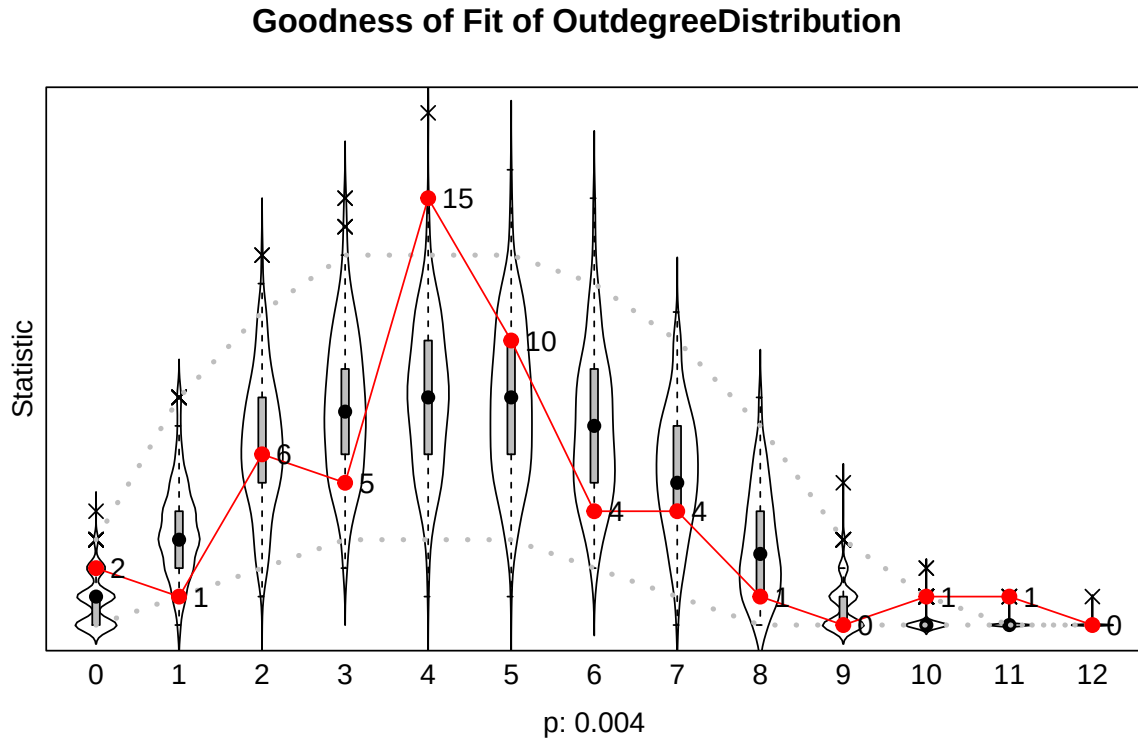
Note: some statistics are not plotted because their variance is 0.
This holds for the statistics: 13 14 15.

10. Discussion and Limits

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We interpret the converged SAOM estimates from **Model 1** (three waves; constant node set). Convergence diagnostics indicate that estimation was successful, so the parameter estimates can be interpreted. In SAOMs, positive parameters increase the attractiveness of creating/maintaining a tie, whereas negative parameters discourage tie formation/maintenance, conditional on the other effects.

Figure 7: Goodness-of-fit for the outdegree distribution (Model 1).



Baseline network tendencies. The baseline outdegree (density) effect is expected to be negative in a sparse nomination network. Reciprocity is typically positive in best-friend nominations; in our model it is estimated as NA (SE = NA), indicating that nominations are substantially more likely when they are mutual.

H1 (Dyadic): sex homophily. The same-sex effect (`sameX` with `sex_cov`) is NA (SE = NA). A positive estimate supports the hypothesis that pupils are more likely to nominate same-sex classmates as best friends, net of other network mechanisms.

H2 (Triadic): transitivity. The transitive triplets effect (`transTrip`) is NA (SE = NA). A positive estimate indicates strong triadic closure: friendship ties tend to form in ways that close transitive structures (“friends of friends become friends”).

H3 (Node-level): sex-based activity. The ego effect of sex (`egoX` with `sex_cov`) is NA (SE = NA). This effect tests whether one sex is systematically more active in *sending* best-friend nominations over time (higher outdegree). The direction and statistical support should be read directly from the estimate and its t-ratio in Table 14.

Structural controls (degree-related). The indegree-related popularity effect (`inPop`) is NA (SE = NA), capturing whether already-popular pupils (high indegree) attract more nominations (positive) or whether nominations avoid already-popular pupils (negative), conditional on the other effects. Outdegree activity (`outAct`) captures heterogeneity in how many nominations pupils send beyond what covariates explain.

Model fit (GoF). Goodness-of-fit compares observed network statistics with simulations from the fitted model. For Model 1, the Monte Carlo p-values are reported in `?@tbl-gof-pvalues-m1`. If any p-values are close to 0 or 1, the model struggles to reproduce that feature. To address misfit, we updated the model once (Model 2) by adding `reciAct` and re-evaluated GoF (see `?@tbl-gof-compare`). The updated model can improve reproduction of reciprocity/triad-related structure, but remaining misfit would motivate trying alternative degree-related effects or adding additional exogenous covariates available in the codebook.

Limitations. First, to satisfy SAOM requirements we restricted the analysis to pupils present in all three waves, reducing the sample and excluding dynamics involving joiners/leavers. Second, best-friend nominations are capped by the questionnaire design, which can affect degree-related mechanisms (e.g., popularity and activity). Third, even after one model update, GoF may indicate remaining discrepancies for some higher-order patterns, suggesting that additional endogenous effects or covariates could further improve fit and substantive explanation.

References

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- Snijders, T. A. B., van de Bunt, G. G., & Steglich, C. E. G. (2010). Introduction to stochastic actor-based models for network dynamics. *Social Networks*, 32(1), 44–60.
- Ripley, R. M., Snijders, T. A. B., Boda, Z., Vörös, A., & Preciado, P. *RSiena Manual* (version matching the installed RSiena package).